Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/05/2011
PI Name:	Oman, Charles M. Ph.D.		
Project Title:	Validation of Assessment Tests and Countermeasures for Detecting and Mitigating Changes in Cognitive Function During Robotics Operations		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Czeisler, Charles (Brigham and Women's Hospital) Liu, Andrew (Massachusetts Institute of Technology) Lockley, Steven (Brigham and Women's Hospital) Barger, Laura (Brigham and Women's Hospital)		
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Task Description:	The goals of this four-year NASA/university project are: (1) characterize the changes in cognitive function during robotic operations that affect performance, (2) validate proxy cognitive and drowsiness assessment tests as predictors of performance changes in a complex operational task, and (3) test the efficacy of fatigue countermeasures (e.g., blue enriched white light, caffeine) to improve cognition during robotic operations. To date, the complex robotics operations aboard the Shuttle and ISS have been successful, but not entirely without incident. Fatigue is common because of chronic sleep deprivation and circadian misalignment of the sleep-wake cycle due to unusual sleep schedules or abrupt "slam shifts" in crew work schedule, and the long duration of robotic operations (up to 6-8 hours). The impairment of cognitive abilities has been identified as a factor leading to errors in performance. In nominal operations, pauses, unnecessary movements and performance inefficiency may reflect cognitive impairment. Preventative countermeasures are necessary to reduce this risk and improve crew safety for current and future robotic operations. This project is a collaborative effort between the MIT Man Vehicle Laboratory (MVL), and the Division of Sleep Medicine at Brigham and Women's Hospital (BWH), with the assistance of collaborators at Jefferson Medical College, NASA Kennedy Space center and the NASA Johnson Space Center Astronaut Office and Mechanical and Robotics Systems Group. Subjects are trained to perform simulated robotics tasks under realistic astronaut schedules. Using a within-subjects design, our experiments investigate how the proposed fatigue countermeasures affect physiologic, cognitive and task performance metrics, thereby enabling their relationship to be understood. If this correlation can be established, it will provide evidence that these metrics could be valuable as indicators of fitness-for-duty. This year, MIT completed development of a virtual reality based Robotic Workstation Simulator, including simulations of representative fly-to/grapple, autosequence, and track-and-capture tasks, and associated performance metrics. BWH and MIT completed protocol development. Subjects are initially consented and screened at BWH then follow a normal sleep schedule (8hr/night) for two to three weeks. During this time, they receive up to six sessions of robotic training as long as they achieve criterion performance during each session. Subjects then shift to a restricted sleep schedule 6 hr/night for one week before admission to the sleep clinic for the inpatient phase of the study, and undergo two refresher training sessions. The inpatient study design consists of four three-day-long experiment blocks. In each block, subjects take a 3 hour nap late on the first day. On the second day they are awoken 9 hours early - a "slam shift" of their sleep schedule - and their performance on robotics, cognitive (Number-Letter, Digit Symbol Substitution), and psychomotor (10 min PVT) tasks is evaluated during a 6 hour robotics session. They will have been awake for 18 hours by the end of the robotics session. Subjects are allowed 8 hours of recovery sleep on the third day, on the preadmission schedule. Physiological measures (plasma melatonin and cortisol, EEG, Optalert eyelid opening speed) and subjective sleepiness (KSS) and alertness measures are taken throughout. The first 3 day block is used as a control: subjects are tested using 90 lux white light. In subsequent blocks, subjects are tested under three conditions with order randomized across subjects: a) 90 lux blue enriched white light, b) 90 lux white light and caffeine (0.3 mg/kg), and c) 90 lux blue enriched light and caffeine. The blue enriched light stimulus is provided by a Solid State Lighting Module similar to that being developed for ISS. So far (5/16/11) BWH and MIT have consented/trained 11 subjects. Three were excluded for medical or personal reasons, and four based on performance in early training sessions. Two have completed testing, and the remainder are completing training and testing. Our goal remains to complete testing of up to 16 more subjects over the next 18 months. Concurrently, MIT is conducting a separate pilot experiment to assess the effect of task type (passive autosequence vs. active track-and-capture) on robotics performance and workload in a dozen subjects who have been awake 18 hours.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An effective, safe, well-tolerated, non-invasive countermeasure for circadian- and fatigue-related deficits in cognition is required for use during in space environments to enhance the safety of crew members. Light exposure has the potential to fulfill this role. Although monochromatic blue light exposure at night has been shown to be most effective at shifting the circadian pacemaker, and improving alertness and performance, these results have not been tested for broadband blue-enriched white light or for light exposure during the biological day. Before light therapy as fatigue countermeasure is operationalized, further research is required to fully understand the effect on complex performance, such as robotics performance. Similarly, while caffeine use is widespread, including on the ISS, uncontrolled use of caffeine may not be optimally timed or of the correct dose to alleviate sleepiness on duty and may interfere with subsequent sleep, thereby increasing fatigue the next day. The current study will investigate the effects of light and caffeine as fatigue countermeasures in a controlled environment with the long-term view to developing specialized countermeasure schedules to maximize alertness and performance of space crew in an environment where even small fatigue-related errors could have catastrophic consequences. These results will also have widespread application on Earth. Approximately 18,000 Air Force pilots and flight crew are exposed to conflicting circadian signals and resulting fatigue from long flights. Tens of thousands of troops are subjected to rapid deployment world-wide and are often required to maintain alertness while suffering from jet lag. Additionally, nearly 180,000 troops are required to be vigilant for more than 24 hours at a time, and thousands of Navy and Coast Guard personnel work for extended periods under circadian misalignment. Over 200,000 commercial airline pilots, flight crews and air traffic controllers are also subjected to extended shifts and circadian misalignment. The NTSB implicated hours of duty and fatigue issues in the 2009 Colgan Air commuter airline crash in Buffalo, NY, and this year the FAA has overhauled rules for air traffic controllers after several highly publicized issues where controllers apparently fell asleep. More than 230,000 ER doctors, staff and medical residents work extended shifts for 30 hours or more. Approximately 1.9 million public safety personnel (police, corrections officers, firefighters, etc.) work extended shifts and are exposed to emergencies which require vigilance for long periods. In total, more than 3 million Americans would directly benefit from the findings of this research. In addition, applications may be possible in standard occupational settings to improve general day-time alertness; in educational settings such as in training sessions and conferences, colleges, and schools where enhanced alertness would assist learning and memory; as a fatigue countermeasure for sleepy car and truck drivers in order to provide them with a short bursts of alertness that will allow them to find a safe place to stop driving and take a break. These wider applications have the potential to reach tens of millions of customers and indeed anyone who could benefit from improved alertness and performance.

Task Progress:	Experiment protocol development and implementation - The experiment protocol was extended from 12 to 13 days to obtain a measurement of subject melatonin levels upon admission to the sleep laboratory. The final selection of cognitive tests was made for the experiment: Visual PVT (10 min), DSST, KDT, PRM, KSS, VAS, Number-letter Test, and PEERS. Two other protocol changes were made in an effort to increase the number of subjects: the age range was widened to 26-55 yoa and the maximum number of training sessions at MIT was increased to 6. We have found that subjects recruited from the general population have required longer periods of training to reach the necessary performance levels than our previous MIT subjects. As of mid May, 2011, we have medically screened 11 volunteers who then began initial robotics training at MIT as a further screening. Six subjects qualified, and two have gone on to complete the entire inpatient phase. A third subject is currently being tested. Hardware and software development - The experimental hardware and software development has been completed and the system successfully used for inpatient experiments. The Robotic Workstation Simulator (RWS-Sim) is built on a mobile cart which has a rack system to mount the 3 LCD monitors and two LED light panels, a desktop for the input devices (hand controllers, keyboard and mouse), and attachments for the CPU and power. An additional light stand holding the two other LED panels is positioned facing the RWS-Sim cart. The experimental software simulates 3 different space teleoperation tasks. For the HTV Track & Capture task, the operator has 90 seconds to maneuver the robotic arm and grapple the free-flying HTV cargo vessel. In each trial, the HTV has a slow translational drift and/or rotation which increase the task difficulty. For the Fly-to & Grapple task, the operator has 5 minutes to move the robot arm through a larger space and successfully grapple a stationary payload, then move it to a specified position near one of the Space Station modules. The final task is the Autosequence Monitoring task in which the operator must first program the arm movements by selecting the designated mode and entering the appropriate data. The subject then monitors the automatic movement of the robot arm and stops the motion if it should move too close to any structures. Operator inputs and simulation state are recorded at 30Hz for later analysis. In addition to these three robotic tasks, the operators must also perform a concurrent secondary task by responding to a flashing alert as quickly as possible. The secondary task is used to provide a measure of the operator's spare workload and monitor any micro-sleep episodes. Outreach: MIT developed a "museum proof" version of the RWS-Sim for use in a new permanent exhibit being developed by the Boston Museum of Science for the Human Life Gallery. The hardware and software are in the third round of testing.
Bibliography Type:	Description: (Last Updated: 12/18/2024)
Abstracts for Journals and Proceedings	Liu AM, Rueger M, Barger LK, Brainard GC, Czeisler CA, Flynn-Evans EE, Forman RE, Lowenthal CS, Natapoff A, Quincy C, Schulz D, Williamson PN, Oman CM, Lockley SW. "Validation of Assessment Tests and Countermeasures for Detecting and Mitigating Changes in Cognitive Function During Robotics Operations." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. Pre-Launch Human Factors, Behavior and Performance Poster Session. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Lowenthal CS, Forman RE, Liu AM, Natapoff A, Oman CM. "Workload measurement during space robotics tasks." 18th IAA Humans in Space Symposium, Houston TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston TX, April 11-15, 2011. Abstract 2304. , Apr-2011
Awards	Czeisler C. "International Academy of Astronautics - elected to full membership, April 2011." Apr-2011
Awards	Oman CM. "International Academy of Astronautics - elected to full membership, April 2010." Apr-2010